

Suppose we need to differentiate a function like

$$f(x) = \sqrt{x^2 + 1}$$

We know how to take the derivative of $x^2 + 1$ and $\sqrt{x}$ individually, but the composition function is more complicated.

It turns out, there is a nice way to take the derivative of composition function

The Chain Rule:

If g is differentiable at x and f is differentiable at $g(x)$, then the composition function $(f \circ g)(x) = f(g(x))$ is differentiable at x and

$$(f \circ g)'(x) = g'(x) f'(g(x))$$

"derivative of inside times derivative of outside"

In other notation,

$$y = f(u), \quad u = g(x), \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Ex: $F(x) = \sqrt{x^2 + 1}$

$$F(x) = (f \circ g)(x)$$

"outside" $f(x) = \sqrt{x} \longrightarrow f'(x) = \frac{1}{2} x^{-1/2}$

"inside" $g(x) = x^2 + 1 \longrightarrow g'(x) = 2x$

$$\begin{aligned} F'(x) &= g'(x) f'(g(x)) \\ &= 2x \cdot \frac{1}{2} (g(x))^{-1/2} \\ &= x \cdot (x^2 + 1)^{-1/2} \\ &= \frac{x}{\sqrt{x^2 + 1}} \end{aligned}$$

Ex: Differentiate $y = \sin(x^2)$

"outside" $f(x) = \sin(x) \longrightarrow \cos(x)$

"inside" $g(x) = x^2 \longrightarrow 2x$

$$y' = g'(x) \cdot f'(g(x)) = 2x \cos(g(x)) = \boxed{2x \cos(x^2)}$$

Ex: Differentiate $y = \sin^2(x)$
 $\hookrightarrow (\sin(x))^2$

"outside" $f(x) = x^2 \longrightarrow f'(x) = 2x$

"inside" $g(x) = \sin(x) \longrightarrow g'(x) = \cos(x)$

$$y' = g'(x) \cdot f'(g(x)) = \cos(x) 2g(x) = \boxed{2\cos(x)\sin(x)}$$

Ex: Using Chain Rule, differentiate $y = \frac{2x}{x^2+1}$

$$y = (2x)(x^2+1)^{-1}$$

$$= 2(x^2+1)^{-1} + (2x) \frac{d}{dx} [(x^2+1)^{-1}]$$

$$= \frac{2}{x^2+1} - \frac{4x^2}{(x^2+1)^2}$$

$$= \frac{2 - 2x^2}{(x^2+1)^2}$$

$$f(x) = x^{-1}, f'(x) = \frac{-1}{x^2}$$

$$g(x) = x^2+1, g'(x) = 2x$$

$$\frac{d}{dx} [(x^2+1)^{-1}] = 2x \frac{-1}{(x^2+1)^2}$$

$\hookrightarrow$ Same answer from quotient Rule

Some compositions are more common than others, so we can calculate their derivatives in general

$$1) F(x) = \sin(g(x))$$

$$F'(x) = g'(x) \cos(g(x))$$

$$2) F(x) = \cos(g(x))$$

$$F'(x) = -g'(x) \sin(g(x))$$

$$3) F(x) = [g(x)]^n \quad \text{for any } n$$

$$F'(x) = n g'(x) [g(x)]^{n-1}$$

Ex: Differentiate $y = (x^3 - 1)^{100}$

$$y' = 100(3x^2)(x^3 - 1)^{99} = \boxed{300x^2(x^3 - 1)^{99}} \quad \text{Using formula above}$$

or

$$f(x) = x^{100}, \quad f'(x) = 100x^{99}$$

$$g(x) = x^3 - 1, \quad g'(x) = 3x^2$$

$$y' = g'(x) f'(g(x)) = 3x^2 \cdot 100(g(x))^{99} = \boxed{300x^2(x^3 - 1)^{99}}$$

Ex: Differentiate $y = \frac{1}{\sqrt[3]{x^2+x+1}}$

$$y = (x^2+x+1)^{-1/3}$$

$$y' = -\frac{1}{3} \overset{\frac{d}{dx}(x^2+x+1)}{(2x+1)} (x^2+x+1)^{-4/3}$$

Ex: Differentiate $g(t) = \left(\frac{t-2}{2t+1}\right)^9$

$$g'(t) = 9 \left(\frac{d}{dt} \frac{t-2}{2t+1}\right) \left(\frac{t-2}{2t+1}\right)^8$$

$$\hookrightarrow \frac{1(2t+1) - (t-2)(2)}{(2t+1)^2} = \frac{2t+1-2t+4}{(2t+1)^2} = \frac{5}{(2t+1)^2}$$

$$g'(t) = 9 \cdot \frac{5}{(2t+1)^2} \left(\frac{t-2}{2t+1}\right)^8 = \frac{45(t-2)^8}{(2t+1)^{10}}$$

or

$$g(t) = \frac{(t-2)^9}{(2t+1)^9} \rightarrow \frac{d}{dt} (t-2)^9 = 9(t-2)^8, \quad \frac{d}{dt} (2t+1)^9 = 18(2t+1)^8$$

$$g'(t) = \frac{(2t+1)^9 \cdot 9(t-2)^8 - (t-2)^9 \cdot 18(2t+1)^8}{(2t+1)^{18}}$$

$$= \frac{[9(2t+1) - 18(t-2)](t-2)^8}{(2t+1)^{10}} = \frac{45(t-2)^8}{(2t+1)^{10}}$$

Ex: Differentiate $F(x) = (2x+1)^5 (x^3-x+1)^4$

$$F'(x) = (x^3-x+1) \frac{d}{dx} (2x+1)^5 + (2x+1) \frac{d}{dx} (x^3-x+1)^4$$

$$= (x^3-x+1)^4 \cdot 5(2)(2x+1)^4 + (2x+1)^5 (3x^2-1)(x^3-x+1)^3$$

$$= 10(x^3-x+1)(2x+1)^4 + 4(2x+1)^5(3x^2-1)(x^3-x+1)^3$$

$$\downarrow = (x^3-x+1)^3 (2x+1)^4 \left(10(x^3-x+1) + 4(2x+1)(3x^2-1) \right)$$

extra
algebra

$$= (x^3-x+1)^3 (2x+1)^4 \left(10x^3 - 10x + 10 + 24x^3 + 12x^2 - 8x - 4 \right)$$

$$= (x^3-x+1)^3 (2x+1)^4 (34x^3 + 12x^2 - 18x + 6)$$

$$= 2(x^3-x+1)^3 (2x+1)^4 (17x^3 + 6x^2 - 9x + 3)$$

We can use the Chain Rule multiple times to differentiate a function with several compositions, for example

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

$$(f \circ g \circ h)'(x) = h'(x) g'(h(x)) f'(g(h(x)))$$

which can be extended to any number of compositions

Ex: If $f(x) = \sin(\cos(\tan(x)))$, find $f'(x)$

$$f'(x) = \cos(\cos(\tan(x))) \frac{d}{dx} \cos(\tan(x))$$

$$= \cos(\cos(\tan(x))) [-\sin(\tan(x)) \sec^2(x)]$$

$$= -\cos(\cos(\tan(x))) \sin(\tan(x)) \sec^2(x)$$

Ex: Differentiate $y = \sqrt{\sec(x^3)}$
 $= (\sec(x^3))^{\frac{1}{2}}$

$$y' = \frac{-1}{2} (\sec(x^3))^{-\frac{1}{2}} \frac{d}{dx} (\sec(x^3))$$

$$= \frac{1}{\sqrt{\sec(x^3)}} \cdot 3x^2 \cdot \sec(x^3) \tan(x^3)$$

$$= \frac{3}{2} x^2 \tan(x^3) \sqrt{\sec(x^3)}$$